

Personality and Performance traits Assessment on Asynchronous Video Interview via Multimodal Fusion

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Abstract

The AVI Challenge 2025 addresses the growing need for robust automated interview systems by introducing two distinct tracks: personality and interview performance assessment. Leveraging a novel dataset comprising asynchronous video interviews (AVIs) from 644 subjects, the challenge aims to advance multimodal machine learning methods for evaluating candidate traits and competencies. Each subject responded to a structured set of generic and personality-specific questions developed using rigorous psychometric and practitioner-based validation. We propose our research results on testing distinct models with all possible modalities combinations with early and late fusion strategies. Our accent was on simplicity of the approach and chosen models meanwhile, enabling to achieve mid-leaderboard positions for both personality and performance tracks. The code can be found in GitHub.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Interview Assessment, Modalities Extraction and Fusion, Emotional AI, Multimodality

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1 Introduction

Automated Video Interviews (AVIs) are increasingly used in personnel selection, yet their validity and fairness rely heavily on advances in multimodal machine learning and robust empirical validation. The AVI Challenge 2025 seeks to drive progress in this area by presenting a comprehensive benchmark for personality and interview performance assessment from AVI data. The challenge consists of two tracks: Track 1 focuses on predicting HEXACO personality traits from candidates' responses to structured personality

questions, while Track 2 expands the task to predicting job-related competencies and overall hireability using all available multimodal information.

The dataset includes interview recordings from 644 applicants responding to six structured questions—two generic and four personality-focused—carefully developed through expert review and practitioner input to maximize psychometric validity. Observer ratings provide ground truth for personality traits, while professional recruiters annotated competencies such as integrity, collegiality, social versatility, development orientation, and overall hireability. By combining structured interview design, and high-quality annotations, the challenge aims to foster research that not only advances technical performance but also addresses critical concerns around reliability, transparency, and bias in automated hiring processes.

2 Related Work

2.1 Emotion Recognition

Through dozens challenges and experiments on emotion recognition framework research field were enriched via leveraging the video modality [14], audio modality [4, 9] starting with Wav2Vec [8] and keeping HuBERT as baseline [15].

2.2 Modalities Fusion

Leveraging multiple modalities representations together is not trivial but with right approach open door to improve metrics. Most of the fusion approaches explore video-audio information fusion [1, 12, 13, 16].

3 Methodology

3.1 Feature Extraction

This section describes the multimodal feature extraction pipeline developed for the AVI Challenge 2025, encompassing meta, video, audio, and text modalities. All modalities except meta were processed using models that showed rich and complementary representations for emotional AI previously which allowed to assume similar high results on predicting personality traits and interview performance.

Meta Modality. Demographic and background information was included to provide relevant context for each subject. The meta modality comprised the subjects' self-reported age, gender, work experience, and education level. Additionally, categorical features indicating race via FairFace [7] (across seven predefined classes) and whether the subject wore glasses [2] during the interview were encoded to capture visual appearance factors that may influence

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perception also gauging possible human assessor biases.

Image Modality. For the video modality, frames were sampled from each AVI at one-second intervals to balance temporal coverage and computational efficiency. Human faces were detected within each frame using a YOLO-face [3]. When a face was present, it was cropped and passed through the EmotiEff [11] to generate frame-level emotional embeddings, which aimed to capture subtle affective cues expressed through facial behavior throughout the interview.

Audio Modality. Raw audio tracks were first extracted and pre-processed to remove background noise and standardize sampling rates. To obtain robust representations of vocal characteristics, each audio segment was embedded using HuBERT [6], a self-supervised speech representation model. These embeddings capture prosodic, phonetic, and paralinguistic information that may convey personality traits and competencies.

Text Modality. To derive semantic features from speech content, each audio track was transcribed using Whisper [10], followed by post-processing to correct for typical errors in longer sequences. The cleaned transcripts were then encoded using emotion english distilroberta [5] embedding model to generate rich, contextualized representations of the spoken responses.

3.2 Modalities Fusion

This subsection compares 2 main approaches of utilizing multi-modal information.

Early Fusion. In this approach, the feature representations extracted from the meta, video, audio, and text modalities were concatenated to form a single multimodal vector for each interview question response. This combined representation preserves the raw signal from each modality and allows the downstream predictive model to learn cross-modal interactions directly during training.

Late Fusion. In contrast, the late fusion strategy processes each modality independently through separate sub-models. The resulting outputs are then averaged to produce the final assessment. This approach maintains the integrity of modality-specific information while allowing the system to balance contributions from each channel, which can be particularly beneficial when certain modalities are missing or noisy.

3.3 Personality Assessment Track

In the first track, the focus is on predicting specific personality traits (Honesty-Humility, Extraversion, Agreeableness and Conscientiousness) from candidates' responses to structured interview questions designed by psychologists to elicit trait-relevant behaviors. Each response is treated as an independent sample in a single-input, single-label regression task.

For this track, each input consists of one video clip corresponding to a single personality question. For example, the file

5484821efdf99b07b28f2300_q3_personality.mp4 represents a candidate's answer to Question 3, which targets the Honesty-Humility trait within the HEXACO model. The ground truth label for this input is a continuous score on a five-point scale derived from expert observer ratings (e.g., 3.5 for Honesty-Humility).

All available modalities—meta, video, audio, and text—are extracted for each response and fused according to the chosen integration strategy (early or late fusion) to produce a trait-specific prediction. The goal of this track is to develop models that can reliably infer nuanced personality characteristics by capturing both the content and delivery of candidate responses.

3.4 Interview Assessment Track

The second track of the AVI Challenge 2025 extends the prediction task to evaluating candidates' overall interview performance through five job-related competencies: Integrity, Collegiality, Social Versatility, Development Orientation, and Overall Hireability. In this multi-input, multi-label regression task, the goal is to assess each competency on a continuous scale from 1 to 5 based on the candidate's responses to all interview questions.

For each subject, the full set of six video responses—including both generic and personality questions—serves as input. For example, the input for a candidate may include: '60fcc84440f8e8c82ca0288_q1_generic.mp4', '60fcc84440f8e8c82ca0288_q2_generic.mp4' and four personality question videos ('q3'- 'q6'). The corresponding ground truth labels are continuous scores for each competency, such as [4, 3.9, 4, 4.3, 4.5].

Features are extracted independently for each response across all modalities (meta, video, audio, and text), then combined using the chosen fusion strategy. In the early fusion approach, all response representations are concatenated into a single input per question per subject before regression. In the late fusion approach, each modality were combined together forming a mulitdimodal representation passed to train regression model. Then each response is processed separately and predictions are aggregated, ensuring that signals from different answers contribute collectively to the final competency scores.

This track reflects a realistic interview scenario where multiple responses inform a holistic evaluation of a candidate's professional competencies, supporting the development of robust, multimodal models for automatic interview performance assessment.

4 Experiment

4.1 Evaluation Metric

The evaluation metric for each personality and performance trait is Mean Squared Error (MSE).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

	Label	Ridge	LinearSVR	CatBoost	MLP (late)	MLP (early)
Personality	Honesty-Humility	0.185	0.187	0.191	0.172	0.171 ✓
	Extraversion	0.237	0.228	0.227	0.161	0.155 ✓
	Agreeableness	0.216	0.215	0.206	0.184 ✓	0.190
	Conscientiousness	0.178	0.176	0.177	0.157	0.150 ✓
Performance	Integrity	0.204	0.209	0.198	0.192 ✓	0.195
	Collegiality	0.281	0.292	0.278	0.284	0.279 ✓
	Social versatility	0.278 ✓	0.291	0.281	0.282	0.284
	Development orientation	0.224	0.216	0.223	0.215 ✓	0.216
	Hireability	0.297 ✓	0.319	0.314	0.317	0.317

Table 1: Modalities Combinations comparison over each trait and each model

4.2 Dataset

Our experiments utilize a multimodal dataset comprising asynchronous video recordings of participants responding to structured interview questions. Each participant answers two generic questions and four personality-related questions, where each personality question is explicitly designed to elicit a specific personality trait.

Ground truth personality trait labels, assigned by professional psychologists, include participant demographics such as age, gender (with options for male, female, non-binary/third gender, or prefer not to say), education level (ranging from less than high school to doctorate), and years of work experience.

The dataset is split at the participant level into training (70%; $n = 450$), validation (10%; $n = 64$), and testing (20%; $n = 130$) subsets, ensuring that videos from the same individual do not appear across splits. The splits maintain consistent distributions of gender, age, and work experience through joint sampling, which supports fair evaluation. While the dataset is used across two experimental tracks — one using only personality question videos and another using all videos — the subject-level split remains identical.

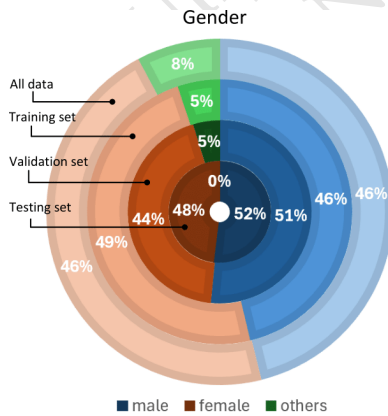


Figure 1: Gender Distribution

4.3 Setting

All models were trained using a batch size of 32 for a total of 150 epochs. We used a fixed learning rate of 0.001 and optimized the models on a single NVIDIA GeForce RTX 3050 Ti GPU with 4GB of VRAM.

5 Results

We evaluated five regression models — Ridge Regression, Linear Support Vector Regression (SVR), CatBoost Regressor, and two Multilayer Perceptrons (MLPs) employing early and late fusion strategies — across all possible modality combinations to identify the optimal configuration for each target personality and performance trait.

5.1 Validation

We conducted an extensive evaluation on the validation split to assess the effectiveness of different models and modality combinations in predicting each target personality and performance trait. Table 1 summarizes the best validation metrics obtained for each trait, model, and corresponding modalities configuration. More on modalities combination illustrated by color in Table 4. Overall, the MLP models consistently outperformed the classical baselines — Ridge Regression, Linear SVR, and CatBoost Regressor — across nearly all traits. Specifically, the MLP with either early fusion achieved the highest scores on most personality traits, demonstrating the advantage of learning joint multimodal representations.

However, we observe that for the traits Social Versatility and Hireability, traditional Ridge Regression slightly outperform the MLPs, suggesting that these particular traits may benefit less from complex feature fusion or that their signal is more readily captured by simpler models. Between the two fusion strategies, early and late fusion exhibit varying performance advantages depending on the trait, highlighting the importance of selecting an appropriate integration scheme for different prediction targets.

5.2 Testing

In this section, we evaluate our model's performance on the official test set. To maximize data utilization and performance, training was performed on the combined train and validation sets before final

testing. This strategy ensures the model benefits from the fullest possible exposure to diverse examples, at the cost of no longer having separate validation feedback — but is standard practice when aiming for leaderboard-competitive results.

Place	Participant	MSE
1	ArchieHe	0.12284
2	Jezoid	0.13724
3	CAS-MAIS	0.14351
4	l_wen	0.14492
5	ABC-Lab	0.16770
6	hdd	0.18909
7	HSEmotion	0.19731
8	abhisheksingh	0.19779
9	nzq	0.20612
10	SonyLai	0.20674
11	YouTu_TX	0.22914
12	xtli	0.23824
13	wjno1	0.24358
14	gkdx2	1.89703

Table 2: Track 1 Leaderboard Results

Place	Participant	MSE
1	ArchieHe	0.18240
2	CAS-MAIS	0.18510
3	ABC-Lab	0.19394
4	l_wen	0.20749
5	HSEmotion	0.22150
6	YouTu_TX	0.24828
7	nzq	0.25540
8	SonyLai	0.25586
9	cherryW	0.37214

Table 3: Track 2 Leaderboard Results

5.3 Discussion

Securing a mid-leaderboard placement (7/14) in Table 2 and (5/9) in Table 3 highlights that our method captures core multimodal signals effectively, yet still falls short of the top-tier entries. The combined training set likely contributed to solid generalizability — no overfitting observed between tracks — but improvements could stem from:

- Exploring enhanced fusion strategies (e.g., cross-attention mechanisms)
- Scaling model capacity in both vertical and horizontal directions
- Accomplishing more analysis following by extended feature engineering

6 Conclusion

The AVI Challenge 2025 illustrates the expanding potential of usage emotional AI prior knowledge, automated video interview assessment, and multimodal human information processing as promising

research frontiers with direct, real-world applications. By integrating structured interview design, rich audiovisual data, and expert-annotated ground truth, this challenge provides a robust testbed for developing systems that can reliably assess personality traits and job-related competencies. As organizations increasingly leverage AVIs and automated assessments to streamline hiring, research in this domain must continue to address challenges related to fairness, transparency, and bias mitigation. Future work should further explore how multimodal models can be made interpretable and generalizable across diverse contexts and populations. Advancing this field responsibly will be essential to ensure that these technologies not only enhance selection processes but also uphold ethical standards and foster trust in AI-driven decision-making.

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Appendix

A Modalities Combinations

All the 4 modalities form 15 possible combinations (skipping combination with no modality). For each modality, the main color was assigned to each modality to make mixing more meaningful. Meta information is associated with black. Video is represented by red tones. Blue is for audio. And, finally, green is for text. The all possible combinations are listed in Table 4.

Color	Meta	Video	Audio	Text
0.000	✓			
0.000		✓		
0.000			✓	
0.000				✓
0.000	✓	✓		
0.000	✓		✓	
0.000	✓			✓
0.000		✓	✓	
0.000		✓		✓
0.000			✓	✓
0.000	✓	✓	✓	
0.000	✓	✓		✓
0.000	✓		✓	✓
0.000		✓	✓	✓
0.000	✓	✓	✓	✓

Table 4: Colors and corresponding modalities combination

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